

## 1. Decision Structures

- a) Display Great job! when grade is 90 or higher:

```
Java
if (grade >= 90) {
    System.out.println("Great job!");
}
```

- b) Display Error when number is less than 20 or greater than 50:

```
Java
if (number < 20 || number > 50) {
    System.out.println("Error");
}
```

- c) Add 2 to the value of y when y is less than 100:

```
Java
if (y < 100) {
    y = y + 2;
}
```

## 6. Evaluating Expressions (Given: size = 100, weight = 50, value = 75)

- a) size > 50 && weight == 50 → **True** (100 is greater than 50, and weight equals 50).
- b) value < 100 && !(weight == 50) → **False** (75 < 100 is true, but !true becomes false).
- c) size >= 100 || value >= 100 → **True** (100 is greater than or equal to 100).
- d) weight < 50 || size < 50 → **False** (Both parts are false).
- e) !(value < 75) → **True** (75 < 75 is false, and !false is true).
- f) !(size > 100 && weight > 50 && value > 75) → **True** (The inside is false because 100 is not greater than 100, and !false is true).
- g) (value < 125 || weight < 76) && size == 100 → **True** (True inside the parentheses, and size equals 100).

## 8. True or False

- a) The condition of an if statement must be a Boolean expression.

- **True** (It must result in either true or false).

**b) A roundoff error can occur when comparing two integers.**

- **False.** Integers are exact whole numbers, so roundoff errors only happen with decimals.

**c) A nested if statement and an if-else if statement are the same.**

- **False.** They are structured differently; nested puts an if inside another if, while if-else if chains them together.

**d) The expression in a switch statement must evaluate to a double.**

- **False.** Switch statements use integers, characters, strings, or enums, not doubles.

**e) Numbers generated by a computer program are actually pseudorandom.**

- **True** (They use a math formula to look random, but follow a pattern).

**f) Specifying a seed value results in a different sequence of "random" numbers each time the program is run.**

- **False.** Using the same seed gives you the exact same sequence every time.

**g) A compound Boolean expression can contain more than two Boolean expressions.**

- **True** (You can link many conditions together using && or ||).

**h) In a logical And expression, both operands must be true for the expression to evaluate to true.**

- **True.**

**i) In logical expressions, && is evaluated before !.**

- **False.** The NOT operator (!) has higher priority and is evaluated first.

**j) The pow( ) method in the Math class is used for exponentiation.**

- **True.**

**k) The statement `x = abs(-3);` will return the value 3.**

- **False.** In Java, it needs to be written as `Math . abs ( -3 )` to work properly without an error.

**I) A diamond shaped object represents a decision in a flowchart.**

- **True.**